

Sujay Korde

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Summary

Second-year B.Tech student specialising in AI & Machine Learning (CGPA: 9.29) with hands-on experience in computer vision, LLM pipelines, causal inference, and full-stack ML system design. Passionate about building intelligent systems that solve real-world problems at scale. Seeking Amazon ML Summer School 2026 to deepen expertise in large-scale machine learning and applied research.

Education

Vishwakarma Institute of Technology, Pune <i>B.Tech – Computer Science & Engineering (AI & Machine Learning)</i> M.G.L Junior College, Paratwada <i>12th Standard (HSC)</i> Avant Grade Public School, Paratwada <i>10th Standard (SSC)</i>	Expected: May 2028 CGPA: 9.29 — SGPA (Latest Sem): 9.38 86.33% 99.40%
Competitive Exams: MHT-CET – 96 %ile JEE – 93.2 %ile	

Projects

- SRE Whisperer** | *FastAPI, Next.js 14, NetworkX, Transformers, SSE* Hackathon Project
- Designed and implemented a full-stack autonomous IT-operations demo system with a **FastAPI** backend featuring causal inference via **NetworkX**, fault simulation, multi-agent orchestration, and live **Server-Sent Events (SSE)** streaming.
 - Built a **Next.js 14** war-room dashboard with real-time metrics, causal dependency graphs, agent-reasoning traces, and human-in-the-loop approval workflows for high-stakes remediation decisions.
 - Optimised transformer models for CPU execution to balance inference performance against resource constraints; delivered deterministic fallback paths, comprehensive test coverage, and reproducible mock environments.
- Nova-Shell** | *C, POSIX, Unix System Calls* Systems Project
- Built a fully functional UNIX shell from scratch in C, implementing process creation with **fork/exec**, I/O redirection, pipeline chaining (**|**) between arbitrary commands, and background process management (**&**).
 - Engineered a custom REPL loop with signal handling (**SIGINT**, **SIGCHLD**), built-in commands (**cd**, **exit**, **history**), and robust error reporting—demonstrating deep knowledge of OS-level process and file-descriptor management.
- WalkNav** | *YOLOv8, OpenCV, Python, Custom Dataset* Assistive AI
- Engineered a real-time assistive navigation system for visually impaired users using **YOLOv8**, achieving **94% mAP50** on a custom-trained outdoor obstacle dataset with sub-100 ms inference latency.
 - Integrated spatial audio feedback with multi-class detection outputs to generate directional alerts, enabling safe pedestrian navigation in dynamic, unstructured environments.
- Protein-Level Disease Prediction via Anomaly Detection** | *scikit-learn, BioPython, Python* Bioinformatics
- Built a scalable bioinformatics pipeline to predict protein-disease associations by extracting residue-level structural features and applying ensemble anomaly-detection models.
 - Achieved **~80% precision** in pathogenicity classification using a tuned Random Forest with automated feature engineering and cross-validated hyperparameter optimisation.

Technical Skills

Languages: Python, C/C++, Java, JavaScript, HTML/CSS, SQL
ML & AI Frameworks: YOLOv8, TensorFlow, scikit-learn, Hugging Face Transformers, LangChain, LlamaIndex
Computer Vision & Data: OpenCV, BioPython, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook
Web & Backend: FastAPI, Next.js 14, React, Streamlit, Server-Sent Events (SSE)
Databases: MongoDB, PostgreSQL, Redis, MySQL
Systems & Infra: POSIX / Unix System Calls, NetworkX, Docker (basic), Git, GitHub
Specialisations: Computer Vision, LLM Pipelines, RAG, Prompt Engineering, Causal Inference, Embedded Systems / IoT

Certifications

Hugging Face – LLM Course (Large Language Models & Transformers)
Google Cloud – Cloud Computing Foundations
Google – Introduction to Generative AI (Google Cloud Skills Boost)
GitHub – GitHub Foundations

Achievements

- Maintained **9.29 CGPA** (9.38 SGPA latest semester) across all 4 semesters at VIT Pune.
- Scored **96th percentile in MHT-CET** and **93.2nd percentile in JEE** among lakhs of engineering aspirants.
- Developed two assistive-technology projects (WalkNav, SRE Whisperer) addressing real-world impact challenges.